



Gennoor Tech
TRAIN. INNOVATE. BUILD.

3-DAY INTENSIVE COURSE · 24 HOURS

Fine-Tuning & Custom LLMs

Customize and train large language models for your use cases

3 24 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

www.gennoor.com · enquiry@gennoor.com

Course Overview

This 3-day intensive program covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python (advanced), PyTorch experience, GPU familiarity.

Who Is This For?



ML Engineers



AI Researchers



L&D Teams
Internal trainers



Technical Leads
Team leads and architects

Learning Outcomes

- 1 **Fine-tune LLMs using** — LoRA, QLoRA, and PEFT techniques
- 2 **Prepare and curate** — training datasets for domain-specific models
- 3 **Evaluate fine-tuned models** — with benchmarks and human evaluation
- 4 **Deploy custom models** — for inference at scale

3-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Fine-Tuning Fundamentals

HANDS-ON LABS + WORKSHOP

- SESSION A
- Fine-Tuning Fundamentals
- When to fine-tune vs prompt engineer vs RAG
 - LoRA, QLoRA, and PEFT — theory and implementation
 - Training data formats: instruction, chat, completion
- SESSION B
- Hands-On & Practice
- Hugging Face Transformers and PEFT libraries
 - Lab: Fine-tune Llama on a custom dataset with QLoRA

02

Dataset Preparation & Training

DEMOS + LABS + DISCUSSION

- SESSION A
- Dataset Preparation
- Data curation: quality, diversity, deduplication
 - Synthetic data generation for fine-tuning
 - Training configuration: learning rate, epochs, batch size
- SESSION B
- Training
- Distributed training with DeepSpeed and FSDP
 - Lab: Prepare and train a domain-specific model

03

Evaluation, Deployment & Optimization

LECTURE + HANDS-ON LABS

- SESSION A
- Evaluation, Deployment
- Evaluation: perplexity, BLEU, human eval, LLM-as-judge
 - Benchmarking against base models
 - ONNX export, quantization (GPTQ, AWQ), vLLM serving
- SESSION B
- Optimization
- Deployment: HF Inference, SageMaker, Azure ML
 - Lab: Deploy and benchmark a fine-tuned model

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the course

- Amazon Lex
- Amazon SageMaker
- Hugging Face

What You Take Home

 **Fine-Tuning Pipeline Template**
Included with course completion

 **Dataset Preparation Scripts**
Included with course completion

 **Evaluation Framework**
Included with course completion

 **Deployment Guide**
Included with course completion

Certification of Completion

 **Gennoor Tech — Certificate of Completion**
Upon completing all 3 days (24 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Fine-Tuning & Custom LLMs". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AI-103
Azure AI Solutions

DP-100
Azure Data Scientist

AI-901
Azure AI Fundamentals (New 2026)

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited
Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.
Builder of open-source MCP servers and Copilot Studio integrations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options ■ **On-Site** — customized at your office | 🖥️ **Virtual Live** — interactive via Teams/Zoom | 🌐 **International** — GCC, Africa, APAC, UK



Ready to Start Your Fine-Tuning & Custom LLMs Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

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